

RESEARCH ARTICLE

Ecofriendly management of anthracnose on avocado caused by *Colletotrichum perseae* in the Mediterranean basin

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Abstract

Background: *Colletotrichum perseae*, recently reported in Europe, causes stem and twig lesions and dieback, and fruit rot on avocado in several countries. Due to EU severe restrictions, there is an urgent need for sustainable alternatives to fungicides. This study aims to evaluate the preventative activity of trade bioformulates against *C. perseae* across detached twig, greenhouse pot-plant grown, and postharvest fruit experiments.

Results: All biological treatments have significantly managed twig anthracnose infections ranging by 37%–91% reductions. *Bacillus amyloliquefaciens* QST713, chitosan, *Trichoderma gamsii* + *Trichoderma asperellum* mixture, and *Equisetum arvense* based bioformulates showed a similar efficacy of the standard fungicide (pyraclostrobin) whilst sweet orange essential oils were less effective. In potted plants grown in greenhouse, *T. asperellum* T34 followed by *T. gamsii* + *T. asperellum* mixture bioformulates revealed the most effective in reducing internal lesions caused by *C. perseae*. On the other hand, bioformulate containing *B. amyloliquefaciens* D747 revealed the most effective treatment in minimizing *C. perseae* anthracnose lesions under post-harvest conditions.

Conclusion: The paper clearly showed that some trade bioformulates could represent a highly effective, environmentally friendly option for managing *C. perseae* of avocado. The implementation of such an approach should be imperative for developing performant strategies for other avocado diseases. This process should begin with preventive applications of *Trichoderma* species as T34 in nursery and preserve the health in postharvest by using *B. amyloliquefaciens* strains-based formulates. Indeed, the strategic adoption of these trade bioalternatives provides a concrete pathway to replace chemical fungicides.

KEYWORDS

anthracnose, avocado, *Bacillus*, biological control agents, sustainable agriculture, *Trichoderma*

INTRODUCTION

Avocado (*Persea americana* Mill.) is a plant native to Mexico and imported by the colonizers in Europe following the discovery of America.¹ In Italy, this crop was first introduced in Sicily (Catania) and then extended into the rest of Southern Italy. The weather conditions of these areas are favorable for this crop to expand but simultaneously

they allowed fungal diseases as anthracnose to determine serious losses. Species belonging to genus *Colletotrichum* are responsible of latent infection that remain quiescent during the fruit development and subsequently lead to fruit rot and stem-end rot² and therefore pose a major concern during the post-harvest stage, minimizing the fruit shelf life.^{3,4} In addition, these fungi represent a threat to young plants, causing apical dieback.^{5,6} Recent studies on the avocado

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plantations in Mediterranean basin has allowed to understand the diversity within *Colletotrichum* species. In Italy the first report of *Colletotrichum fructicola* and *Colletotrichum gloeosporioides* causing branch canker was in 2016 combined with *Botryosphaeriaceae* and *Diaporthe* spp.⁵ and just a few years later *Colletotrichum fioriniae* was reported as a causal agent of stem end-rot on fruit.⁷ In 2024, *Colletotrichum perseae* and *C. gloeosporioides* s.s. infections have been associated with stem lesions, dieback, and fruit rot, clearly indicating the emergence of complex disease syndromes affecting both nursery and field plants.⁸

For a long period, the use of fungicides as a means for controlling anthracnose seemed to be the only option available for avocado farmers. Nowadays, the number of chemical molecules to control fungal diseases has been strongly reduced, opening the way for a sustainable approach that preserves both the environment and consumer health. The frequent use of fungicides has promoted the development of resistance in fungal pathogens and cross-resistance among pathogenic microorganisms, as well as the contamination of agro-ecosystems, causing serious and irreversible environmental pollution.⁹

Consequently, it has become necessary to find alternatives for disease control in pre- and post-harvest stages.¹⁰ For this reason, new commercial antagonistic-based bioformulations or products suitable for organic cultivation have come to play a pivotal role.^{11,12}

Scientific interests in the biological management of avocado anthracnose, caused by *Colletotrichum* spp., have increased significantly. Nowadays, the main objective is to find ecofriendly substitutes to synthetic chemical fungicides, particularly for the critical post-harvest stage.

Substances derived from organic products offer an alternative to the development of natural fungicides, as they represent a selective method that is safer for humans and the environment.¹³ These products can act against the pathogen through various modes of action, such as inducing systemic host resistance and exerting a direct effect versus the fungal mycelium growth. Some studies clearly showed the performance of bacteria colonizing the fruit surface that can reduce the spore germination of fungi, as it was reported for *Kosakonia cowanii* VG1² and lactic acid bacteria strains.¹⁴ Meanwhile, biological control agents such as fungi compete against the pathogen for space and nutrients. Among these, one of the most used in biological control is *Trichoderma* genus, which includes few species with the most antagonistic ability.¹⁵

Despite these advances, the application of biological control in Mediterranean avocado systems remains limited and requires further investigation. Factors such as climatic variability, pathogen diversity, and host-microbe interactions can influence the efficacy of biocontrol agents under field conditions. Therefore, integrating biological control into sustainable disease management programs requires a deeper understanding of pathogen ecology, microbial interactions, and environmental influences specific to Mediterranean regions.

To guarantee optimal cultivation, it is crucial to guarantee that planting stock origin is disease-free. Consequently, correct placement of treatment on young avocado plantation during critical stages, for example, as pruning, assumed a key role as a preventative measure

against disease. This could ensure healthy of plant which will grow and bear fruit in the best conditions.

In light of the above considerations, this study investigated (i) the antifungal activity of some trade bio-formulates against *C. perseae* on detached young twigs (ii) the in vivo bioformulates effectiveness on avocado plantation and (iii) their relative postharvest performance in reducing fruit anthracnose.

MATERIALS AND METHODS

Fungal isolates and bioformulates

Three experiments were performed to evaluate the performance of several trade bioformulates. The first on detached avocado twigs under laboratory conditions; the second on potted grown plants in a greenhouse nursery and the last on avocado fruit under postharvest conditions. For all experiments *C. perseae* APIN16 was chosen and used for the inoculation.⁸ This isolate belongs to fungal collection of the Laboratory of the Department of Agriculture, Food and Environment, University of Catania, isolated from young plant of cv. Hass grafted on 'Day rootstock' in 2024 in the eastern of Sicily. The samples showed stem lesions, wood discoloration and dieback, often starting at the grafting point and causing the death of the plants.⁸

In detail, Table 1 summarizes the active ingredients and bioformulates tested in the experiments. Since they were conducted on detached avocado twigs, plants, and fruits under controlled conditions, not all trade products were used in every trial. The treatment selection varied according to the type of assay and the fixed objectives. Among the tested bioformulates, only Remedier®, Serenade ASO®, and Prevatect® were included in all the experiments. These products were selected because of their high potential biocontrol activity and different modes of action. Remedier® contains *Trichoderma gamsii* 080 and *Trichoderma harzianum* 012 (modes of action: on competition for space and nutrients, mycoparasitism, antibiosis and induced systemic resistance), Serenade ASO® includes *Bacillus amyloliquefaciens* QST713 (modes of action: direct action, induction of systemic acquired resistance and competition for space and nutrients) while Prevatect® contains chitosan, a biopolymer known for its ability to stimulate plant defense responses.¹⁶ Furthermore, Cabrio WG® (a.i. pyraclostrobin) was included as a standard fungicide in the detached assay, along with Prev-am Plus®, based on sweet orange essential oil and Equiset®, containing *Equisetum arvense* extract. For the field and fruit experiments, however, the following T34 Biocontrol®, based on *Trichoderma asperellum* T34 and Amylo-X®, contains *B. amyloliquefaciens* strain D747 were evaluated.

Detached twig assay

At the end of 2024 detached avocado assays were conducted to evaluate the efficacy of biological control agents. Twigs (cv Hass) with a diameter ranging from 0.9 to 2 cm and approximately 25 cm long

TABLE 1 Bioformulates used for trials, applied at the rates recommended by the manufacturers.

Active ingredient	Trade name	Manufacturer	Rate
Pyraclostrobin	Cabrio WG [®]	BASF Italia S.p.A.	112.5 g/hL
Chitosan	Prevatect ^{®a}	Ascenza Italia S.r.l.	4 L/hL
<i>Bacillus amyloliquefaciens</i> QST713	Serenade ASO ^{®a}	Bayer Crop Science S.r.l., Milano, Italy	0.8 L/hL
Sweet orange essential oil 5.88%	Prev-am Plus [®]	Ascenza Italia S.r.l.	800 mL/hL
<i>Trichoderma gamsii</i> 080 + <i>Trichoderma asperellum</i> 012	Remedier ^{®a}	Gowan Italia S.r.l.	250 g/hL
<i>Equisetum arvense</i>	Equiset [®]	Ascenza Italia S.r.l.	500 mL/hL
<i>Trichoderma asperellum</i> T34	T34 Biocontrol [®]	Cifo S.r.l.	25 g/hL
<i>Bacillus amyloliquefaciens</i> D747	Amylo-X [®]	CBC (Europe) S.r.l. – Biogard, Brussels-Belgium	250 g/hL

^aProducts used in all the experiments.

were selected. Trade bioformulates tested herein were Prevatect[®], Serenade ASO[®], Prev-am Plus[®], Remedier[®] and Equiset[®] and a fungicide standard (Cabrio WG[®]) (Table 1) to reduce the lesions caused by *C. perseae*. The treatments were performed preventively 72 h before the inoculation of pathogens. For the experimental scheme, six branches per treatment were used for each timing. The experiment was carried out twice. First of all, branches were sterilized for 10 min with 1% hypochlorite (NaClO), then rinsed twice with sterile distilled water (SDW) and left to dry. After disinfection, they were then placed vertically in polystyrene containers filled with perlite (inert), and cuts were made to simulate the pruning of branches. The treatments were carried out by spraying until runoff (about 100 mL) into the cut wound. The positive control consists of a water treatment and then inoculated.

Successively, each branch was inoculated with 10 µL conidial suspension (1×10^6 Colony Forming Units, CFU) of *C. perseae* (APIN 16) in the wound with pipette and covered with Parafilm[®] (American National Can). The containers were placed in a growth chamber with a 12 h photoperiod at $25 \pm 1^\circ\text{C}$. The reduction in lesion length (cm) compared to the positive control was measured using destructive method, assessing the internal lesion of the branch 21 days after inoculation.

Greenhouse experiment on pot grown avocado plant

Further investigations were subsequently carried out directly in a greenhouse to test the effectiveness of products in field conditions. The experiment was conducted during May 2025, using the bioformulates most effective in the detached assays. In addition, two formulations have also been added based on *B. amyloliquefaciens* and *Trichoderma* spp., and testing whether there is any difference in the use of different strains. Five commercial organic products (Prevatect[®], Serenade ASO[®], Amylo-X[®], Remedier[®], T34 Biocontrol[®]) were therefore used for this experiment to evaluate their preventive efficacy on the field. For the scheme of the trial, 4-year-old plants of avocado cv. 'Hass' grafted on Duke 7, located in a greenhouse (temperature about 22–32°C and relative humidity about 85%) in Malaga (South of Spain). Three avocado plants per treatment were used for each timing with three points of inoculation for each plant.

TABLE 2 Analysis of variance for degree of internal lesion based on average lesion size (cm) incited by *Colletotrichum perseae* APIN16 on avocado cultivar Hass.

Factor(s)	Internal lesion (cm) ^a		
	df	F	p
Treatment	6	45.8661	<0.0001
Assay	1	0.0179	0.893857 ^{ns}
Treatment × assay	6	0.0343	0.999819 ^{ns}

Note: The *p* values (< 0.0001) indicate high significance for treatment factor.

Abbreviations: df, degrees of freedom; ns, not significant data.

^aF test of fixed effects, df, and *p* value associated to F.

The treatments were applied with the dosage indicated on Table 1 and were performed 72 h before and at the same time of the pathogen inoculation.

To evaluate the treatments, 6-mm-diameter wounds were produced on the avocado fruit with a cork borer. The assigned treatments were immediately applied to the wounds, which were then allowed to absorb for 30 min before inoculation. Thereafter, the lesions were covered with Parafilm[®] to prevent excessive desiccation during the period following the inoculation.

After 72 h the Parafilm[®] was removed, the wounds were inoculated using a conidial suspension (1×10^6 CFU) of *C. perseae* and to preserve humidity, Parafilm[®] was re-applied. The positive control consists of the same amount of SDW used for treatment. After 30 days, the external lesions were measured in order to detect the twigs and check whether the infection began in the untreated control.

Experiment on avocado fruit in postharvest

In vivo experiment was performed to determine the most effective bioformulates against anthracnose fruit infection caused by *C. perseae* under postharvest conditions. To this aim, the same bioformulates used in previous experiment were tested on avocado fruits cv. 'Hass' (Table 1). The treatments were performed 96 h before and at the same time of the fungal inoculation.

For this experimental scheme, five avocado fruit per treatment were used for each timing, with two points of inoculation in the equatorial area of each fruit. Healthy avocado fruit cultivar ‘Hass’ were selected at the same ripening stage (i.e., unripe). The fruit surface was at first disinfected with a solution containing 70% water, 20% commercial bleach (NaClO), and 10% ethanol (96%), rinsed in SDW twice

TABLE 3 Post-hoc analyses about main effects of performance of biological treatments in reducing infections in detached twig assay of avocado caused by *Colletotrichum perseeae* APIN16.

Treatment	Average internal lesion (cm) ^a	Reduction (%)
Pyraclostrobin	0.32 ± 0.05c	91.62
<i>Bacillus amyloliquefaciens</i> QST 713	0.36 ± 0.07c	90.58
Chitosan	0.45 ± 0.10c	88.22
<i>Trichoderma gamsii</i> 080 + <i>T. asperellum</i> 012	0.52 ± 0.10c	86.39
<i>Equisetum arvense</i>	0.53 ± 0.06c	86.13
Sweet orange essential oil (a.i. 5.88%)	1.41 ± 0.30b	63.09
Untreated and inoculated control	3.82 ± 0.58a	-

^aCombined data from two assays ± standard error of the mean (SEM). Means for each data were obtained from six replicates (avocado Hass twigs). According to Fisher's least significance difference test, values followed by different letters within the column are significantly different ($\alpha = 0.05$).

for 4 min, and air dried on sterile paper. After that, avocado fruits were wounded used a sterile needle of 0.5 mm and immersed in 3 liters of the bioformulate suspension for 10 min at the label rate (Table 1). The fruit was treated with the biological product 3 days before and at the same time of fungal inoculation. For the positive control, the fruit were simply immersed in tap water. Successively, each fruit was inoculated with 10 µL of conidial suspension (1×10^6 CFU) of *C. perseeae*. After they dried out, the fruit were placed in plastic containers and incubated in a growth chamber with a 12-h photoperiod at about $25 \pm 1^\circ\text{C}$ and about 70% relative humidity. The test was carried out twice. The mean lesion size (two perpendicular axes or diameters) was measured after 5 days of incubation.

Data analysis

Data regarding performance evaluation of biological treatments in reducing *C. perseeae* infections through detached twigs assay, greenhouse and postharvest experiments on avocado fruit cv. ‘Hass’ were analyzed by using the Statistica package software (vers. 10; Statsoft Inc., Tulsa, OK, USA). The arithmetic means of lesions were calculated averaging the values for all replicates of each treatment. Initial analyses of disease lesion on twigs and fruit were performed by calculating *F* and *p* values associated to evaluate whether the effects of single factor and interactions are significant (two-way ANOVA). In the main effect posthoc analyses, the mean values of twig lesion and fruit infections values were subsequently separated by the Fisher's least significance difference test ($p = 0.05$).



FIGURE 1 Exemplificative view about effects of bioformulates on avocado Hass twigs against *Colletotrichum perseeae*. (a) Positive control; (b) Sweet orange essential oil; (c) *Equisetum arvense*; (d) *Trichoderma gamsii* 080 + *Trichoderma asperellum* 012; (e) Chitosan; (f) *Bacillus amyloliquefaciens* QST 713; (g) Pyraclostrobin.

RESULTS AND DISCUSSION

Detached twig assay

A two-way ANOVA was preliminarily performed for detached assays under controlled environment about performance of bioformulates in controlling *C. perseae* twig infections (Table 2). These data clearly showed that the treatment factor had a significant effect on reduction of twig lesion caused by fungus, whereas the assay and treatment $\times$ assay interaction were not significant.

Comprehensively, these data showed that the ranking of effectiveness is similar through the trials and therefore the data are highly reproducible.

TABLE 4 Post-hoc analyses of performance of biological treatments in reducing infections on avocado plants cv Hass greenhouse experiment caused by *Colletotrichum perseae*.

Treatment	Average internal lesion (cm) ^a	Reduction (%)
<i>Trichoderma asperellum</i> T34	1.27 $\pm$ 0.30c	76.78
<i>Trichoderma gamsii</i> 080 + <i>T. asperellum</i> 012	1.58 $\pm$ 0.28bc	71.12
<i>Bacillus amyloliquefaciens</i> D747	1.72 $\pm$ 0.25bc	68.56
<i>B. amyloliquefaciens</i> QST713	1.83 $\pm$ 0.18bc	66.54
Chitosan	2.14 $\pm$ 0.28b	60.88
Untreated and inoculated control	5.47 $\pm$ 0.69a	-

^aData from single experiment $\pm$ SEM (standard error of the mean). Means for each data were obtained from three replicates (avocado plants), each formed by three branches. According to Fisher's least significance differences test, values followed by different letters within the column are significantly different ($\alpha = 0.05$).

As a consequence of previous ANOVA, the repeated assays were combined and subjected to post-hoc analysis of the main effects in reducing fungal lesion (Table 3, Figure 1a–g).

Based on internal lesion size, all bioformulates were able to significantly reduce infections caused by *C. perseae* on detached avocado twigs highlighting few significant differences among treatments. Indeed, except for sweet orange essential oil which showed least effectiveness, the remaining bioformulates and fungicide (a.i. pyraclostrobin) provided similar performances in controlling fungal disease under operative conditions and after 21 days of incubation (Figure 1a–g). These preliminary findings cannot be considered definitive since they need to validate through in field experiment as consistently reported in the literature.^{17,18} So these preliminary data have stimulated the next in field experiment focusing on the most effective bioformulates, that is, BCAs and chitosan and including new trade *Trichoderma* and *Bacillus*-based bioformulates since they have provided among the highest disease reductions.

Greenhouse experiment on pot-grown avocado plants

In the greenhouse experiment, located at Malaga experimental station IHSM-La Mayora in Spain and carried out by using 4-year-old potted grown plants, all treatments provided once again significant reductions of internal lesion caused by *C. perseae* if compared to the untreated control (Table 4, Figure 2a–f).

The highest performances were detected for the bioformulate containing *T. asperellum* T34 (Figure 2f), which reported the highest percentage disease reduction in field (almost 77%). This is due to the fact that these BCA species not only reduce infection but is also capable of promoting plant growth (Sokra 2024).¹⁹ Lower reductions (but not significant) were recorded for formulates containing *T. asperellum* + *T. gamsii*



FIGURE 2 Effects of biological products on avocado plant cv. Hass against *Colletotrichum perseae*. (a) Positive control; (b) Chitosan; (c) *Bacillus amyloliquefaciens* QST 713; (d) *Bacillus amyloliquefaciens* strain D747; (e) *Trichoderma gamsii* 080 + *Trichoderma asperellum* 012; (f) *Trichoderma asperellum* T34.

mixture, *B. amyloliquefaciens* D747, and *B. amyloliquefaciens* QST713. Notably, the worst performance has been detected for chitosan based formulate that differed significantly from T34 but not from the remaining bioformulates. These last data about polymer are partially divergent with the previous assay obtained under laboratory conditions but it probably due to different operative conditions. Unlike to detached assays for testing BCAs, additional factors could have played a crucial role in field experiment as consistently corroborated by past reports about anthracnose on tropical crops.²⁰

Postharvest fruit experiment

As first experiment, a significant effect in reducing fungal decay on avocado fruit was detected under operative postharvest conditions only for treatment factor whereas two-way ANOVA did not reveal any significance for trial and treatment $\times$ trial interaction thus indicating a high reproducibility of experiment (Table 5, Figure 3a–f).

Consequently, data from two trials were combined and subjected to post-hoc main effects (Table 6).

TABLE 5 Analysis of variance for degree of fruit infections based on average decay size (cm) incited by *Colletotrichum perseeae* APIN16 on avocado fruit cultivar Hass in repeated experiments.

Factor(s)	Fruit decay (cm) ^a		
	df	F	p
Treatment	5	93.748	<0.0001
Trial	1	0.089	0.766633 ^{ns}
Treatment $\times$ trial	5	0.222	0.952223 ^{ns}

Note: The *p* values (< 0.0001) indicate high significance for treatment factor.

Abbreviations: df, degrees of freedom; ns, not significant data.

^aF test of fixed effects, df, and *p* value associated to F.

Based on these data, all treatments provided significant fruit decay reductions if compared to relative control. In detail, *B. amyloliquefaciens* D747 bioformulate was the most effective since it provided the highest percentage reduction (about 85%), significantly different from the remaining bioformulates (Figure 3f). *B. amyloliquefaciens* QST713, *T. asperellum* T34 and *T. asperellum* ICC 012 + *T. gamsii* ICC 080 mixture bioformulates achieved good and similar performance in reducing fruit infections whereas chitosan once again provided the worst efficacy confirming data obtained in greenhouse experiment. Another noteworthy aspect concerns the reversal in the trend regarding BCA effectiveness reported herein, with a commercial *Bacillus* based bioformulation showing the highest efficacy, unlike previous detached twig assays, in which a commercial *Trichoderma* based bioformulation showed the best efficacy. This observation should not come as a complete surprise, since other research on mango fruit rot caused by *Neofusicoccum* likewise reported higher efficacy of a *Bacillus*-based

TABLE 6 Post-hoc analyses of performance of biological treatments in reducing infections on avocado fruit cv Hass assay caused by *Colletotrichum perseeae*.

Treatment	Average size decay (cm) ^a	Reduction (%)
<i>Bacillus amyloliquefaciens</i> D747	0.32 $\pm$ 0.09d	84.90
<i>B. amyloliquefaciens</i> QST713	0.81 $\pm$ 0.07c	61.79
<i>Trichoderma asperellum</i> T34	0.82 $\pm$ 0.10c	61.32
<i>Trichoderma asperellum</i> 012 + <i>T. gamsii</i> 080	0.91 $\pm$ 0.05bc	57.07
Chitosan	1.06 $\pm$ 0.05b	50.00
Untreated and inoculated control	2.12 $\pm$ 0.12a	-

^aCombined data from two assays $\pm$ standard error of the mean (SEM).

Means for each data were obtained from five replicates (avocado fruit with two infection wounds). According to Fisher's least significance difference test, values followed by different letters within the column are significantly different ($\alpha = 0.05$).



FIGURE 3 Postharvest effects of biological products on avocado fruit cv Hass against *Colletotrichum perseeae*. (a) Positive control; (b) Chitosan; (c) *Trichoderma gamsii* 080 + *Trichoderma asperellum* 012; (d) *Trichoderma asperellum* T34; (e) *Bacillus amyloliquefaciens* QST 713; (f) *Bacillus amyloliquefaciens* D747.

bioformulate if compared to *Trichoderma* ones.²¹ This confirms once again the ability of *B. amyloliquefaciens* species to produce antifungal mechanisms, lipopeptides and volatile organic compounds, thus proposing it as an ideal competitor for nutrients, and a disease resistance inducer.^{22,23}

Comprehensively, our findings provide a clear framework for discriminating among trade BCAs to be used against *C. perseeae* reported herein as causal agent of twig, branch and fruit infections, since it is well known in literature that their efficacy depends not only on operative conditions (in field or in post-harvest), targeted fungi and tropical host plant but also on involved canopy organs.^{17,18,20,24,25}

CONCLUSIONS

In light of the above overall findings, BCAs tested herein represent an alternative to fungicides highly effective for preventing anthracnose infections caused by *C. perseeae*, which was currently reported as a serious threat for avocado for the first time in Europe. This study provides a valuable benchmark against which the currently information and future findings obtained under different greenhouse/field and postharvest operative conditions can be compared. The strategic placement of BCAs offers a concrete pathway to reduce/eliminate fungicide dependency, aligning Mediterranean agriculture with modern environmental standards.^{26,27}

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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